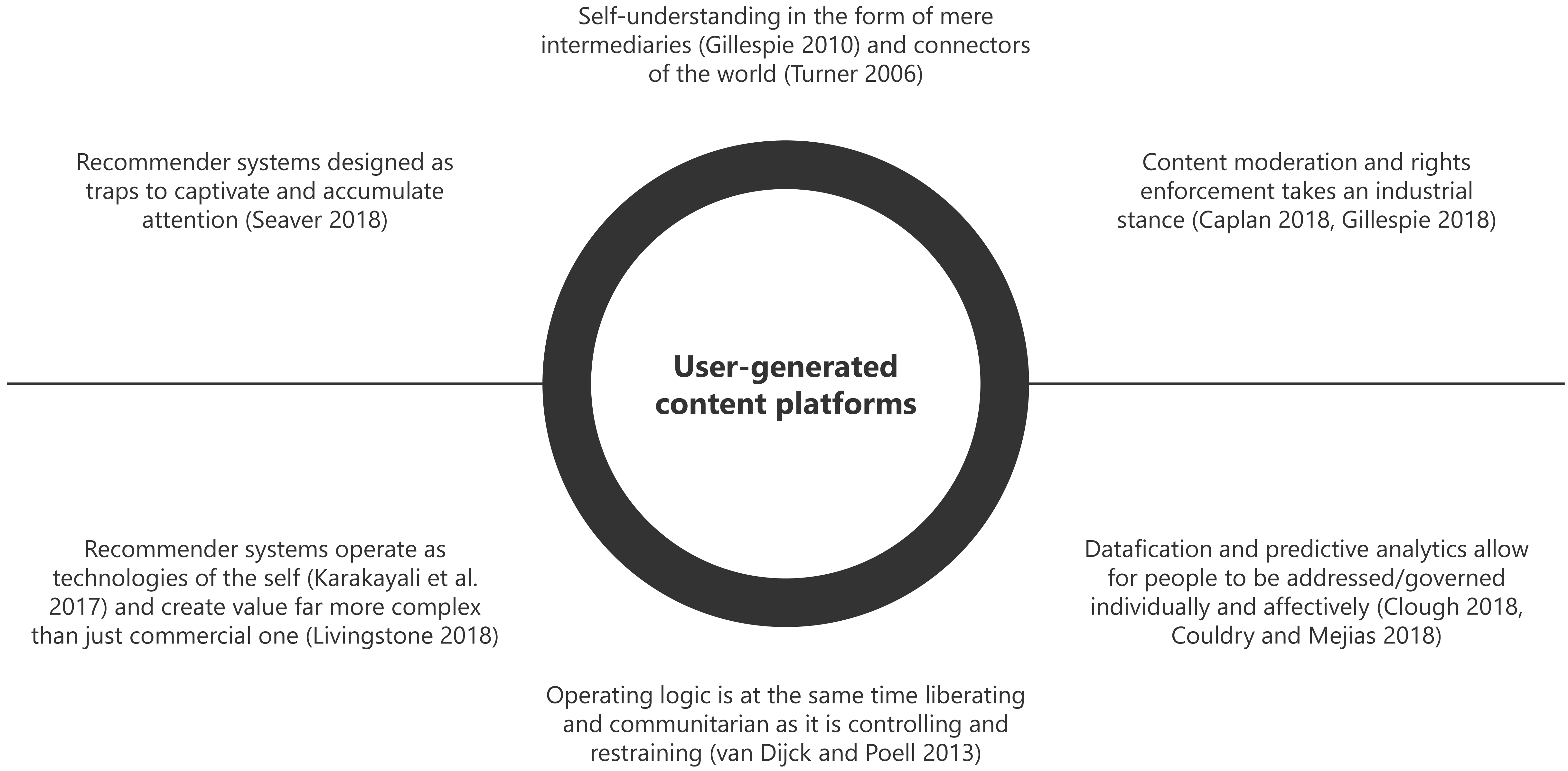
Reconstructing Digital Culture

Methods to understand interactions between online communities, media organizations, and algorithmic recommender systems

THE UNIVERSITY of EDINBURGH

Andreas Schellewald, University of Edinburgh
a.o.schellewald@sms.ed.ac.uk


Self-understanding in the form of mere intermediaries (Gillespie 2010) and connectors of the world (Turner 2006)
Recommender systems designed as traps to captivate and accumulate attention (Seaver 2018)
Content moderation and rights enforcement takes an industrial stance (Caplan 2018, Gillespie 2018)
User-generated content platforms
Recommender systems operate as technologies of the self (Karakayali et al. 2017) and create value far more complex than just commercial one (Livingstone 2018)
Datafication and predictive analytics allow for people to be addressed/governed individually and affectively (Clough 2018, Couldry and Mejias 2018)
Operating logic is at the same time liberating and communitarian as it is controlling and restraining (van Dijck and Poell 2013)

Three challenges of studying algorithms (Kitchin 2016)

Access

- Not only hard to get but data will shaped through technical decisions made while accessing
- Institutional inequality between academia and industry (boyd and Crawford 2012)
- Transparency - knowing comes from seeing - forms ill-fitting metaphor (Ananny and Crawford 2016)

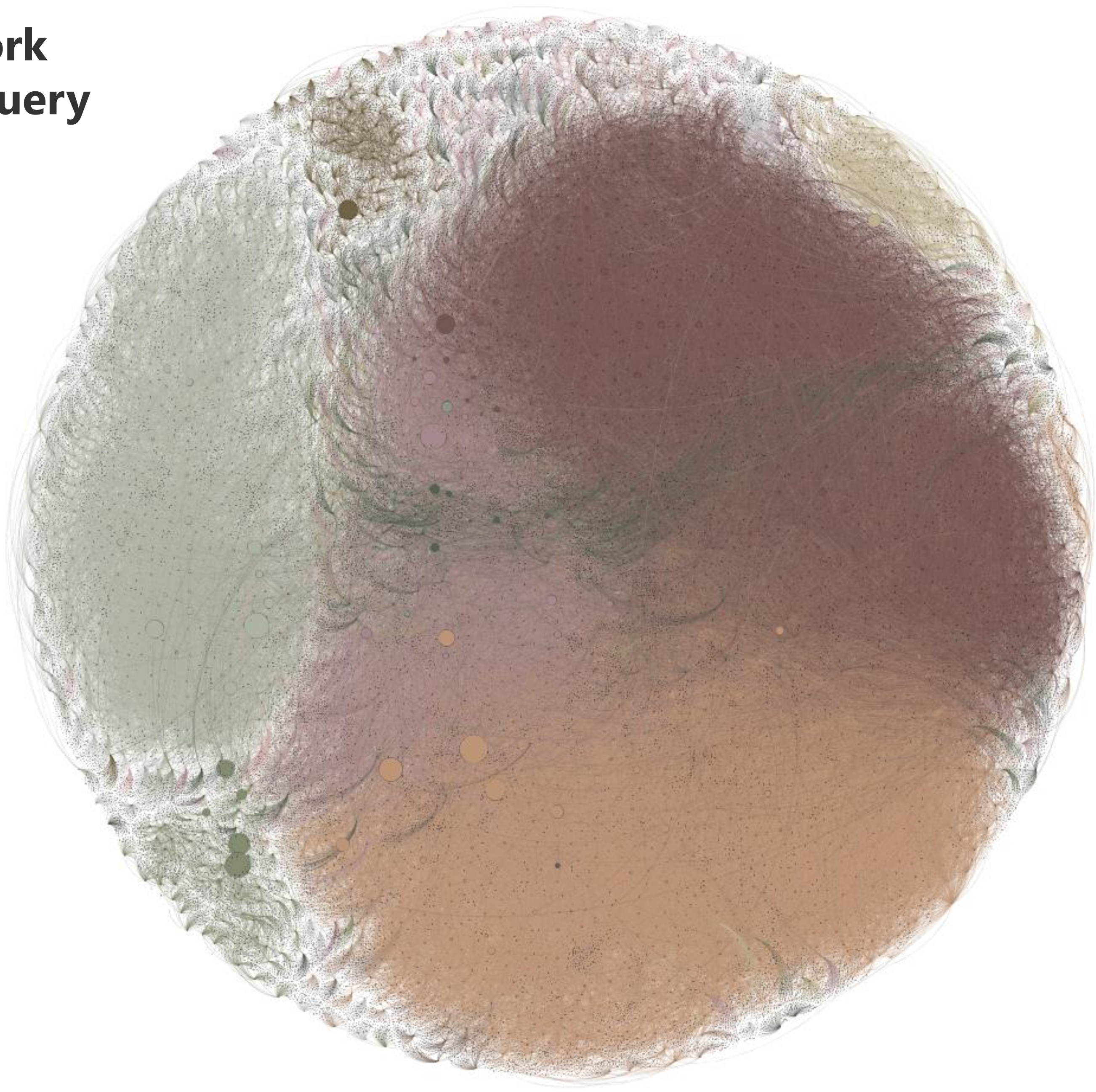
Embeddedness

- Algorithms are part of larger, complex nettings of different algorithms, datasets, protocols, standards, laws, and so forth
- Techno-centric inquiry – code with consequences – must but supplemented a reflection of social constructedness and institutional embeddedness (Gillespie 2014)

Contingency

- Algorithms are constantly worked on, unfold and interact in dynamic and recursive ways
- Not only must different approaches be combined but also discrepancies between qualitative and quantitative bridged (Halford et al. 2014)

Related videos network for YouTube search query “got house stark”



Nodes	36972
Edges	123310
Clustering coefficient	.11
Average degree	6.8
Diameter	17
Range clustering coefficient within clusters	.1 / .15
Clusters / colour by	Louvain Modularity
Node size by	Betweenness
Layout	ForceAtlas2

Reconstructing YouTube recommendations

Data gathering

A related video network is the 'backbone' dataset upon which YouTube builds recommendations and personalization.

- Gives us an idea of what the contents on YouTube look like broad variety of users.
- YouTube Data v3 API (public access).
- Top seeds from the search query (same results as when using native search via web interface).
- Related videos info for depths 0 and 1 (meaning get related videos for initial seeds, depth 0 and 1, for which again get the related videos data).
- Limited official technical literature available (for instance Baluja et al. 2008, Davidson et al. 2010, Bendersky et al. 2014, Covington et al. 2016) but enough to engage in the task of reconstructing with some epistemic certainty.

Network analysis

Cluster through Louvain community detection algorithm (Blondel et al. 2008)

- Assumption is that YouTube creates relations based on similarity (Baluja et al 2008, Davidson et al 2010, Simonet 2013, Bendersky et al 2014).
- Louvain algorithm generates clusters by optimizing communities towards densely connected nodes.
- Commonly used and reliable community detection method to gain first insight into composition of large networks (Lancichinetti and Fortunato 2011, Papadopoulos et al. 2011, Orman et al. 2012).

Force-directed layout with scaled gravity

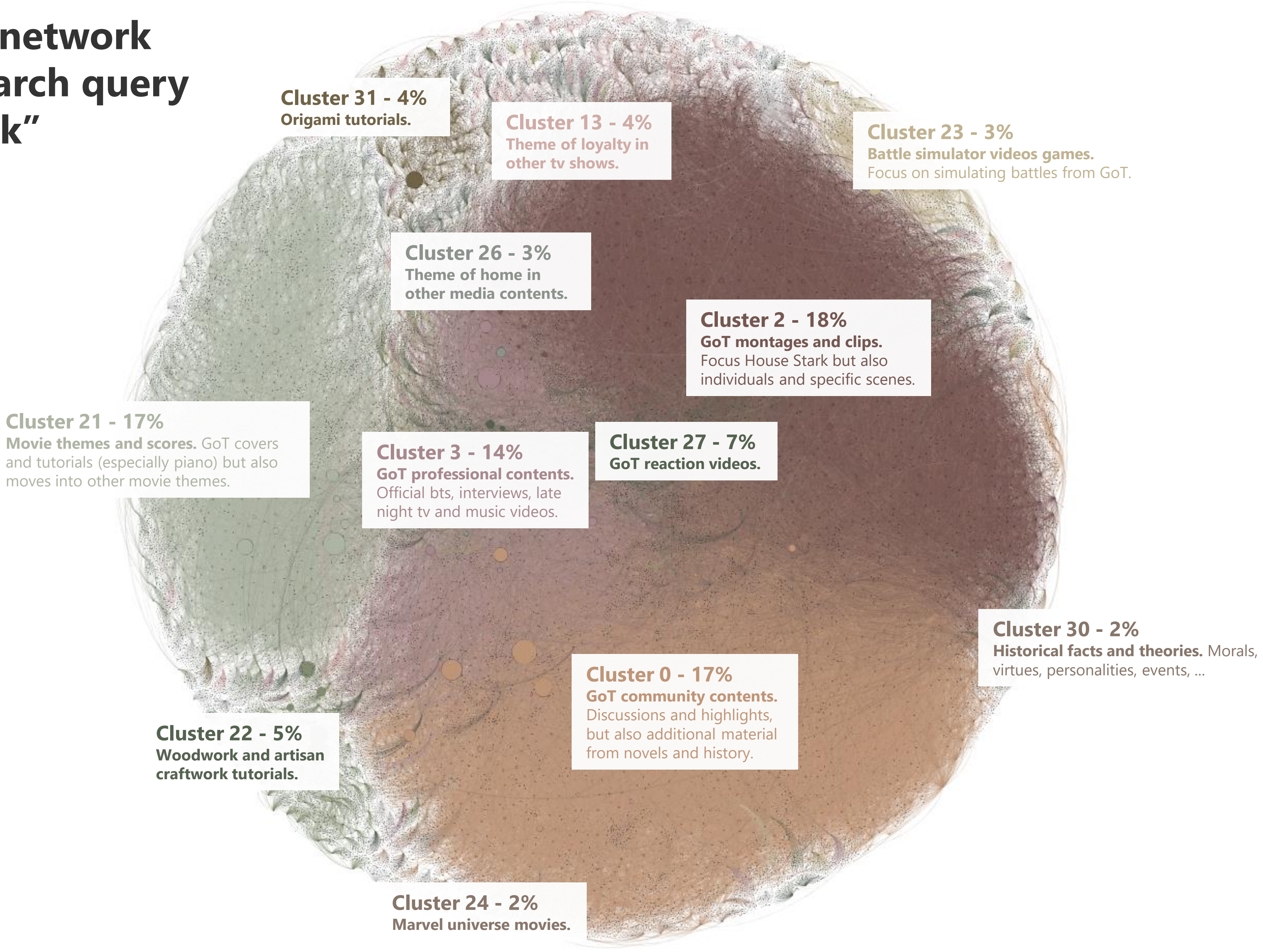
- Central nodes are placed centrally, cluster behave accordingly, gravity scaling enhances this by more compact and comparable spatialization (Bannister et al. 2013).

Content analysis

Computationally supported content analysis as analysis of network culture (DiMaggio et al. 2013, Fuhse and Mützel 2011, Mützel 2015, Lewis et al. 2013, Lucas et al. 2015)

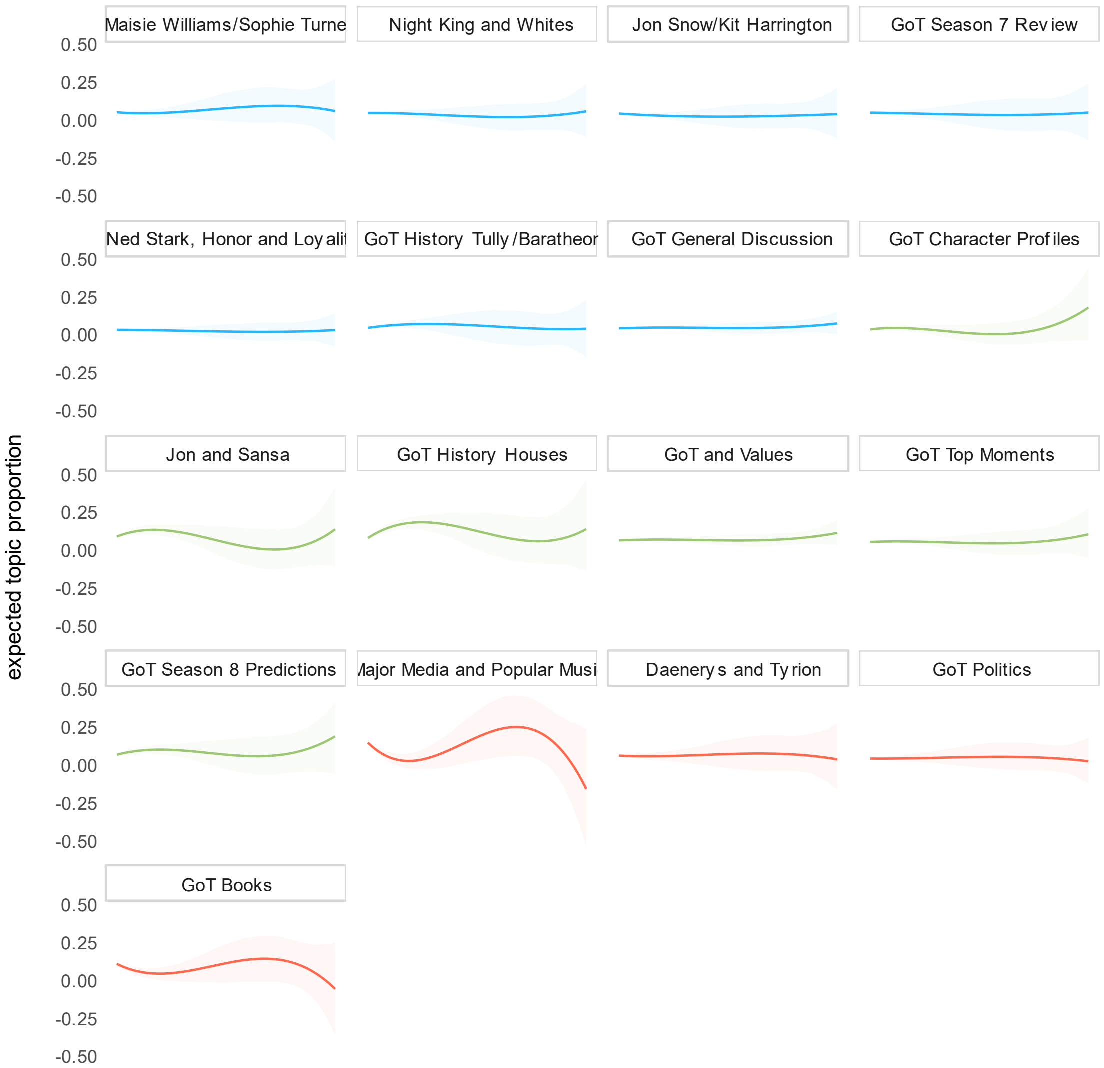
- Assessing themes of clusters through analysis of video titles and video tags (most operable considering size).
- (1) Inductive categorization of lower and upper bound of central (betweenness centrality) and influential nodes (view count and engagement ratio).
- (2) Extraction of central terms through analysis of co-occurrences (Deerwester et al. 1990, Lund and Burgess 1996).
- (3) Structural topic modelling for each corpus of video titles with centrality and influence measures as covariates (Roberts et al. 2013)

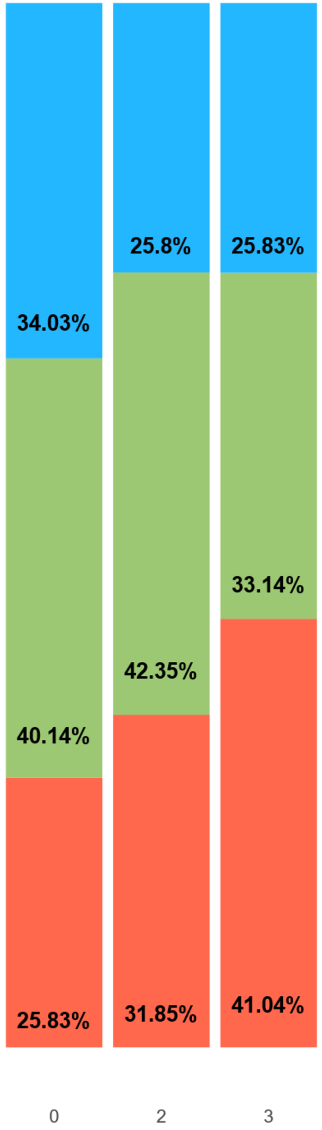
Related videos network for YouTube search query “got house stark”



Nodes	36972
Edges	123310
Clustering coefficient	.11
Average degree	6.8
Diameter	17
Range clustering coefficient within clusters	.1 / .15
Clusters / colour by	Louvain Modularity
Node size by	Betweenness
Layout	ForceAtlas2

Effect of betweenness centrality on topic distribution for Clusters 2, 3, and 0





Topics placed **neutral** by the algorithm are general discussion of GoT but also videos featuring the cast and interviews with them.

Topics **pushed** by the algorithm are especially montages of GoT characters and community videos concerning the upcoming seasons.

Topics **slowed down** by the algorithm especially are major media and popular music contents but also videos discussing the GoT books.

However, note that these videos still make up 1/3 and more of the traffic!

These methods **allows us to understand and reconstruct the linked ecologies** (Abbott 2005) which are at and can be **at work in contemporary digital culture**. An analysis that is not final but rather sets the ground to start tracing further.

However, we have to **emphasise the organizational role algorithms play** as material entities equipped with agency herein. They observe, classify, and order, are adapted and changed. Yet most importantly, algorithms are implemented on servers, have a runtime operation which allows them to **create affective presences on the platform**.

Thanks!

a.o.schellewald@sms.ed.ac.uk

@qonto / soi.hypotheses.org

References

Abbot, A. (2005). Linked Ecologies: States and Universities as Environments for Professions. In Sociological Theory, 23(3), 245-274.

Ananny, M. and Crawford, K. (2016). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. In New Media & Society, 1-17.

Baluja, S. et al. (2008). Video Suggestion and Discovery for YouTube. Taking Random Walks Through the View Graph. In Proceedings of WWW 2008, Beijing, China.

Bannister, M.J. et al. (2013). Force-Directed Graph Drawing Using Social Gravity and Scaling. In W. Didimo and M. Patrignani eds., Graph Drawing. GD 2012. Lecture Notes in Computer Science, vol 7704. Berlin: Springer, 414-425.

Bendersky, M. et al. (2014). Up Next: Retrieval Methods for Large Scale Related Video Suggestion. In Proceedings of KDD 2014, New York, US.

Blondel, V.D. et al. (2008). Fast unfolding of communities in large networks. In Journal of Statistical Mechanics, P10008.

boyd, d. and Crawford, K. (2012). Critical Questions for Big Data. Provocations for a cultural, technological, and scholarly phenomenon. In Information, Communication & Society, 15(5), 662-679.

Caplan, R. (2018). Content or Context Moderation? Artisanal, Community-Reliant, and Industrial Approaches. New York: Data & Society.

Clough, P. (2018). The User Unconscious. On Affect, Media, and Measure. Minneapolis: University of Minnesota Press.

Couldry, N. and Mejias, U.A. (2018). Data Colonialism: Rethinking Big Data's Relation to the Contemporary Subject. In Television & New Media, 1-14.

Covington, P. et al. (2016). Deep Neural Networks for YouTube Recommendations. In Proceedings of RecSys 2016, Boston, US.

Davidson, J. et al. (2010). The YouTube Video Recommendation System. In Proceedings of RecSys 2010, Barcelona, Spain.

Deerwester, S.C. et al (1990). Indexing by latent semantic analysis. In Journal of the American Society for Information Science, 41(6), 391–407.

DiMaggio, P. et al. (2013). Exploiting affinities between topic modelling and the sociological perspective on culture. In Poetics, 41, 570-606.

Fuhse, J. and Mützel, S. (2011). Tackling connections, structure, and meaning in networks: quantitative and qualitative methods in sociological network research. In Qual Quant, 45, 1067-1089.

Gillespie, T. (2010). The politics of platforms. In New Media and Society, 12(3), 347-364.

Gillespie, T. (2014). The Relevance of Algorithms. In T. Gillespie, P.J. Boczkowski, and K.A. Foot eds., Media Technologies: Essays on Communication, Materiality, and Society. Cambridge, MA: MIT Press, 167-193.

Gillespie, T. (2018). Custodians of the Internet. Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media. New Haven: Yale University Press.

Halford, S. et al. (2014). Big Data: Methodological Challenges and Approaches for Sociological Analysis. In Sociology, 48(4), 663-681.

Karakayali, N., Kostem, B., and Galip, I. (2017). Recommendation Systems as Technologies of the Self: Algorithmic Control and the Formation of Music Taste. In Theory, Culture and Society, 35(2), 3-24.

Kitchin, R. (2017). Thinking critically about and researching algorithms. In Information, Communication & Society, 20(1), 14-29.

Lancichinetti, A. and Fortunato, S. (2011). Limits of modularity maximization in community detection. In Physical Review, 84, 066122.

Lewis, S.C. et al. (2013). Content Analysis in an Era of Big Data A Hybrid Approach to Computational and Manual Methods. In Journal of Broadcasting & Electronic Media, 57(1), 34-52.

Livingstone, S. (2018). Audiences in an Age of Datafication: Critical Questions for Media Research. In Television & New Media, 1-14.

Lucas, C. et al. (2015). Computer-Assisted Text Analysis for Comparative Politics. In Political Analysis, 25, 254-277.

Lund, K. and Burgess, C. (1996). Producing high-dimensional semantic spaces from lexical co-occurrence. In Behavior Research Methods, Instruments, and Computers, 28 (2), 203– 208.

Miheleca, R. and Tarau, P. (2004). TextRank Brining Order into Texts. In Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing.

Mützel, S. (2015). Facing Big Data: Making sociology relevant. In Big Data & Society, 1-4.

Orman, G.K. et al. (2012). Comparative evaluation of community detection algorithms: a topological approach. In Journal of Statistical Mechanics, P08001.

Papadopoulos, S. et al. (2011). Community detection in Social Media. Performance and application considerations. In Data Mining and Knowledge Discovery, 24, 515-554.

Roberts, M.E. et al. (2013). The structural topic model and applied social science. In Advances in Neural Information Processing Systems Workshop on Topic Models: Computation, Application, and Evaluation.

Seaver, N. (2018). Captivating algorithms: Recommender systems as traps. In Journal of Material Culture, 1-16.

Turner, F. (2006). From Counterculture to Cyberculture. Stewart Brand, the Whole Earth Network, and the Rise of Digital Utopianism. Chicago: University of Chicago Press.

van Dijck, J. and Poell, T. (2013). Understanding Social Media Logic. In Media and Communication, 1(1), 2-14.